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SkySense-O: Towards Open-World Remote Sensing Interpretation with Vision-Centric Visual-Language Modeling

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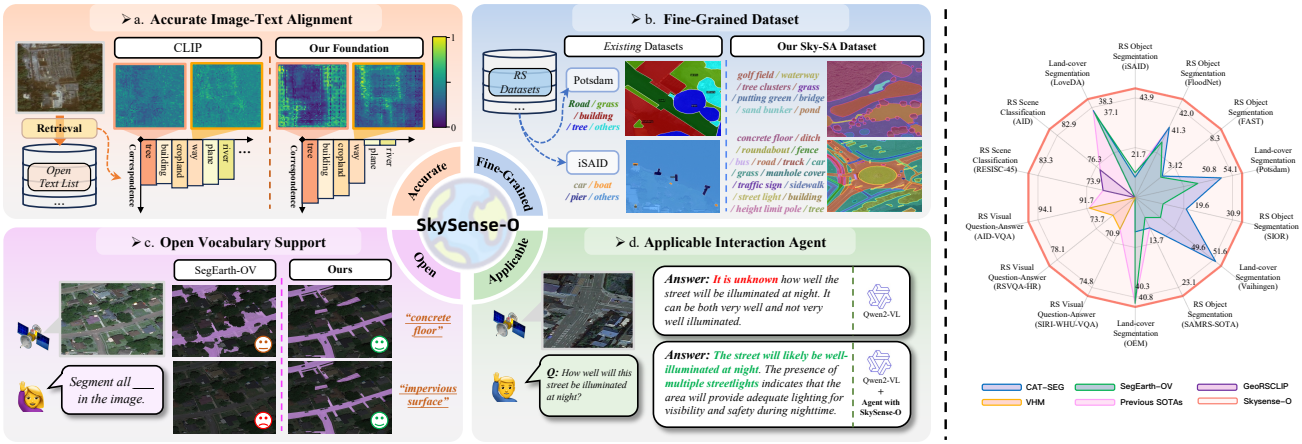


Figure 1. SkySense-O offers 4 key advantages from the foundation model to the agent application (Left), leading to state-of-the-art zero-shot performances across 14 downstream datasets over 4 distinct RS interpretation tasks (Right)

Abstract

Open-world interpretation aims to accurately localize and recognize all objects within images by vision-language models (VLMs). While substantial progress has been made in this task for natural images, the advancements for remote sensing (RS) images still remain limited, primarily due to these two challenges. 1) Existing RS semantic categories are limited, particularly for pixel-level interpretation datasets. 2) Distinguishing among diverse RS spatial regions solely by language space is challenging due to the dense and intricate spatial distribution in open-world RS imagery. To address the first issue, we develop a fine-grained RS interpretation dataset, Sky-SA, which contains 183,375 high-quality local image-text pairs with full-pixel manual annotations, covering 1,763 category labels, exhibiting richer semantics and higher density than previous datasets. Afterwards, to solve the second issue, we introduce the vision-centric principle for vision-language modeling. Specifically, in the pre-training stage, the visual self-supervised paradigm is incorporated into image-text alignment, reducing the degradation of general visual representation capabilities of existing paradigms. Then, we construct a visual-relevance knowledge graph across open-category texts and further develop a novel vision-centric image-text contrastive loss for fine-tuning with text prompts. This new model, denoted as SkySense-O, demonstrates impressive zero-shot capabilities on a thorough evaluation encompassing 14 datasets over 4 tasks, from recognizing to reasoning and classification to localization. Specifically, it outperforms the latest models such as SegEarth-OV, GeoRSCLIP, and VHM by a large margin, i.e., 11.95%, 8.04% and 3.55% on average respectively. The code is publicly available to facilitate further research at https://github.com/zqcrafts/SkySense-O.

1. Introduction

Remote sensing interpretation (RSI) is a critical and practical task in Earth observation, aiming to identify the semantic types of specific ground objects in remote sensing (RS) images (such as segmentation [1–4], detection [5–10], change detection [11, 12], and visual question answering

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(VQA) [13–15]). Early works typically involved annotating RS images with fixed categories and exploring RSI through supervised learning methods. These approaches do not generalize well to unseen new categories in open-world scenarios, thus limiting their practical applicability. Moreover, as RS images becoming more accessible than ever before, there is an increasing demand for open-world interpretations with a broad category scope. Hence, there is a soaring interest in exploring a comprehensive open-world remote sensing interpretation model for many scenarios and tasks [16, 17].

Challenges. In the domain of natural image scenes, significant progress has been achieved in open-world interpretation tasks through large-scale localization and recognition datasets like AS-1B [18] and COCO-Stuff [19], combined with contrastive language-image pretrained model (CLIP) shown in Fig. 1a. However, the open-world RS interpretation model is still out of reach due to following two reasons: *1)* From the dataset perspective, the existing annotations either exhibit sparse mask labels or lack diversity in category descriptions (see Fig.1b). *2)* From the modeling perspective, the language model inherently encompasses strong contextual dependence [20], making it challenging to directly use as prompts for decoupling common RS objects and intertwined land-cover categories in open-world.

Sky-SA Dataset. For Struggle *(1)*, inspired by SAM [21] and ASM [18] in the natural image domain, we propose the Sky-SA dataset with the largest number of category labels for open-world RS interpretation, including 1,763 category texts from 183,375 mask-text pairs. The Sky-SA is mainly characterized by its *text openness* and *fine granularity* (Fig. 1b). Specifically, different from general semantic segmentation datasets, the categories of Sky-SA are annotated by short semantic tags in a completely open manner. In other words, the annotators are doing fill-in-the-blank rather than multiple-choice. This enables the model to learn pixel-level open-world interpretation akin to human cognition [18] (Fig. 1c). Furthermore, similar to COCO-Stuff [19], all recognizable pixels of Sky-SA are densely annotated, which further enhances the semantic openness and the contextual relationships from wide semantic categories.

Vision-Centric Modeling. For Struggle *(2)*, motivated by [22], we introduce the vision-centric principle to correct the potential alignment ambiguity arising from language guidance. Firstly, we initialize the visual encoder from SkySense [23], an advanced RS visual foundation model, and then collect large-scale RS image-caption datasets for pre-training, which mainly include SkyScript [23], RS5M [24] and OSM [25]. Motivated by DetailCLIP [26], to mitigate the degradation of general visual representation capabilities originating from SkySense, we maintain the visual contrastive loss and masked modeling loss during the pre-training process of image-text alignment. Secondly, for fine-tuning region-level alignment, we leverage GPT-4V [27] to construct a visual-centric knowledge graph across all categories of the Sky-SA dataset, where the node of the graph represents each category, and the edge weight indicates the visual similarity scores between each pair of categories. Using the graph, we derive a vision-centric regularization that focuses on distinguishing categories exhibiting low visual similarity but strong text-level contextual relationships in RS images. Finally, we explore the impacts of Skysense-O as agent on MLLM models (Fig. 1d). In summary, to achieve open-world RS interpretation model, SkySense-O is meticulously developed from dataset to modeling and from pre-training to fine-tuning (Fig. 2), contributions of this paper include:

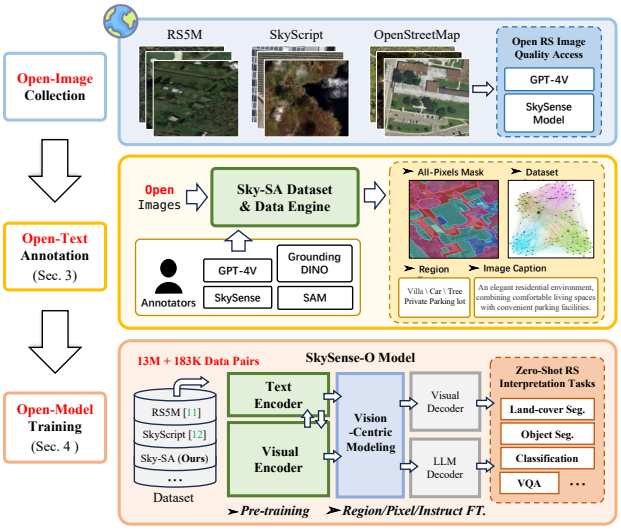


Figure 2. The flowchart of the overall SkySense-O development.

- We develop Sky-SA dataset with large-scale and high-quality annotated texts for open-world RS interpretation.
- The vision-centric principle is first introduced into RS visual-language modeling, primarily instantiated into training paradigm, vision-centric graph, and vision-centric regularization from pre-training to fine-tuning.
- We propose SkySense-O, a remarkable open-world RS interpretation model, which achieves SOTA across 14 zero-shot downstream datasets as shown in Fig. 1.

2. Related work

2.1. Remote Sensing Interpretation

Remote Sensing (RS) interpretation aims to accurately identify and analyze ground objects from RS data, with applications in environmental monitoring, urban planning, and agricultural surveillance [35–42]. Early deep learning approaches focused on closed-set supervised models, in-

Dataset	Spatial Granularity			Open	Dense
	Image	*Region*	*Pixel*	Text	Annotation
Natural Image					
SA-1B [21]	✓	✓		✓	
COCO-Stuff [19]		✓	✓		✓
Remote Sensing Image					
AID [28], RESISC [29]	✓				
OpenStreetMap [30]		✓	✓		
RS5M [24], SkyScript [31]	✓			✓	
Potsdam, LoveDA [32]		✓	✓		✓
Valiggan, OEM [30]		✓	✓		✓
iSAID [33], SIOR [34]		✓	✓		
FAST [34], SOTA [34]		✓	✓		
Sky-SA (ours)	✓	✓	✓	✓	✓

Table 1. **Property comparisons** with popular vision-language datasets. The proposed Sky-SA dataset is characterized by its semantic openness, high-density and fine-granularity.



(a) Florence-2 (b) SAM (c) Grounding DINO-1.6 **(d) Sky-SA Dataset**

Figure 3. **Visual comparisons** between the predictions of existing open-world models and the annotations of our Sky-SA dataset.

cluding image classification [43, 44], object detection [45–49], change detection [11, 12], and semantic segmentation [28, 50–56]. Recently, CLIP [57] and Vision Language Models (VLMs) have enabled the transition from closed-set to open-set interpretation, allowing models to recognize broader categories [17, 58]. Additionally, models like GPT-4 [59] and LLaVA [60] have facilitated the development of instruction-tuning datasets [61], enhancing the flexibility of RS models in human-machine interactions.

2.2. Vision Foundation Models for Remote Sensing

Vision foundation models are instrumental in advancing RS interpretation toward open-set capabilities through two primary pre-training paradigms, vision-only self-supervised and vision-language self-supervised [22, 43, 62–67] paradigms. Vision-only self-supervised models utilize large-scale unannotated RS images to develop robust visual representations for various tasks [23, 68–71]. For example, SkySense [23] integrates multiple modalities and geo-contextual information within a contrastive framework, while Masked Image Modeling (MIM) methods like RingMo [72] and Scale-MAE [73] are proposed for further enhancing visual feature learning. Vision-language self-supervised models originate from CLIP. Recently some efforts such as RemoteCLIP [74], RS5M [24], and SkyScript [23] have customized for the RS domain using extensive RS image-caption pairs.

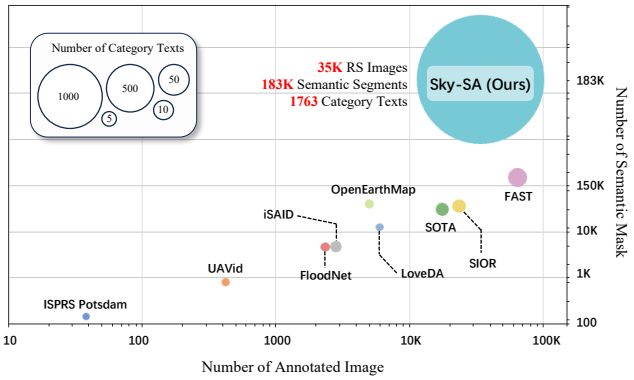


Figure 4. **Scale comparisons** with existing RS semantic segmentation datasets. We see that the proposed Sky-SA dataset has a significantly larger category scale and diversity than prior datasets.

3. Sky-SA Dataset

3.1. Motivation

Recently, image-text alignment models based on natural images, such as the open-vocabulary methods [75] and MLLMs like LLaVA [60], have achieved remarkable success. This is primarily due to well-constructed datasets in the natural image domain compared to the RS image domain. There have been numerous high-quality datasets with rich categories in the natural image community, such as COCO-Stuff [19] and ADE [76]. However, as shown in Tab.1 and Fig.4, such resources remain scarce in the RS community, which significantly limits the development of open-world RS interpretation tasks. Moreover, existing open-world models, such as SAM [21], Florence-2 [77], and Grounding-DINO [78], are also unable to achieve dense and fine-grained interpretation on remote sensing images as shown in Fig. 3. To relieve this issue, we design a data engine to help annotators more efficiently and accurately perform full-pixel annotation on unlabeled RS images, leading to the creation of the Sky-SA dataset as following section.

3.2. Data Engine

Open-Images Collection. In the standard collection process of a closed-set dataset, a category list is predefined, and images that do not contain any items from this list will be filtered out. This manner might lead to dataset bias [79] in remote sensing images. Unlike the close-set methods, we compile a comprehensive collection of images from various available RS datasets. To minimize redundancy in manual annotation, we leverage the SkySense model [23], which possesses image-level visual discrimination capabilities for RS images, to filter visually redundant images.

Open-Text Fine-Grained Annotation. In the annotation process, drawing inspiration from the popular COCO-Stuff [19] dataset, we prioritize fine-grained semantics and

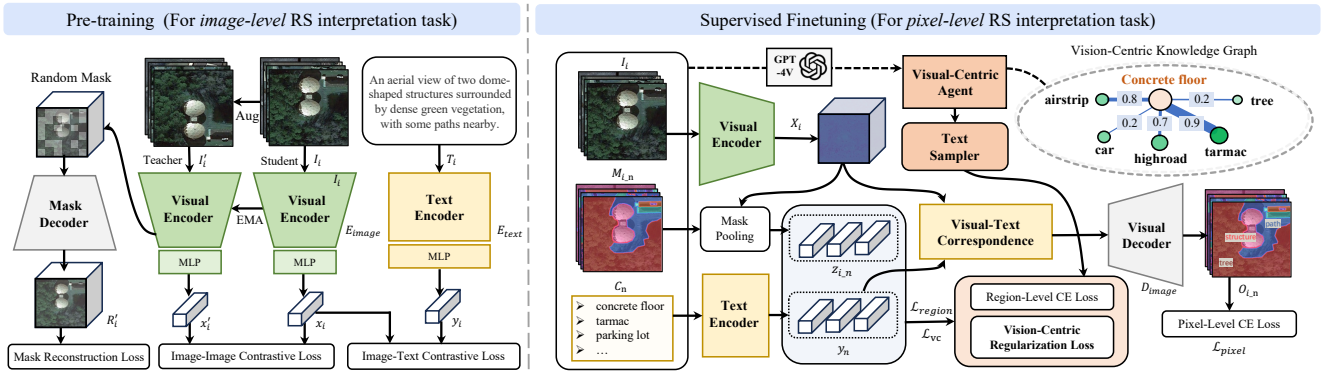


Figure 5. **The framework of SkySense-O**, where training strategy includes pre-training and fine-tuning stage for open-world interpretation.

full-pixel annotation as core labeling principles. Specifically, there are two key points: 1) the dataset encompasses a rich variety of semantic categories; 2) almost all pixels in each image need to be annotated. The former aids in activating the broad semantic awareness of the vision-language pretrained model, while the latter helps establish comprehensive connections among local semantics throughout each image. Furthermore, to capture a broader semantic range, we employ a short-caption labeling manner for each region, which does not constrain the annotation labels. Additionally, to improve annotation efficiency, we first utilize GPT-4V [27] to generate the image content, and then employ Grounding-DINO [78] and SAM [21] for pre-annotation, allowing annotators to make careful corrections based on these preliminary labels.

Dataset Statistics. Finally, we develop a high-quality pixel-level remote sensing interpretation dataset, named Sky-SA. As shown in Fig. 4, the Sky-SA dataset consists of 35,000 RS images, mainly including 1,763 category texts from 183,375 segment-text pairs (encompassing approximately 500 distinct categories of land-covering and RS objects after synonym filtering), which far surpasses existing remote sensing segmentation datasets. Moreover, Sky-SA is also the first pixel-level open-text annotated dataset in the field of remote sensing (see Fig. 1), laying a solid data foundation for the development of open-world RS interpretation.

4. SkySense-O Model

4.1. Pretraining for Foundation Model

The training of SkySense-O is primarily divided into pre-training and fine-tuning stages as shown in Fig. 5. During pre-training, given the input image I_i and its corresponding text T_i, for $i = 1, ..., B$, where B denotes mini-batch size, the model mainly consists of an image encoder E_{img} and a text encoder E_{text}. To maintain the model's visual perception capabilities [26], especially for densely distributed objects during the image-text alignment process, the vision encoder is initialized by SkySense [23] and its visual self-supervised paradigms are still maintained, *i.e.* masked image modeling and image contrastive learning. Given $x = E_{img}(I)$, $y = E_{text}(T)$, mask decoder output R', the overall pretraining loss is computed as Eq. 1:

$$\mathcal{L}_{pretrain} = CE\,(x, y) + CE\,(x, x') + MR\,(I', R')\,, \quad (1)$$

where I' and x' are the augmented version of I and x respectively, MR is mask reconstruction loss following [71] and CE denotes cross-entropy loss computing as:

$$CE(x, y) = -\frac{1}{B}\sum_{i=1}^{B}\log\frac{\exp(sim(x_i, y_i)/\tau)}{\sum_{j=1}^{B}\exp(sim(x_i, y_j)/\tau)}, \quad (2)$$

where τ is logit temperature and sim is cosine similarity.

4.2. Finetuning for Interpretation Model

Region-level Finetuning. During the region-level fine-tuning stage, based on the proposed Sky-SA dataset, given a set of candidate category texts C_n and masks M_n for each image, where $n = 1, ..., N$, we apply mask pooling to extract local features for each category from the images, and then compute the region-level cross-entropy loss with the corresponding category, which can be expressed as:

$$\mathcal{L}_{region} = \sum_{n=1}^{N} CE\big(Pooling\,(x_n, M_n)\,, E_{text}(C_n)\big). \quad (3)$$

Pixel-level Finetuning. To achieve pixel-level interpretation, referring to the existing open-vocabulary semantic segmentation model [80], we first compute the similarity between the image encoder and the text encoder at the token level to obtain coarse location features. Subsequently, a visual decoder D_{img} is employed to up-sample these features to the fine-grained segmentation results, constrained by a pixel-wise cross-entropy loss as Eq. 4:

$$\mathcal{L}_{pixel} = \sum_{n=1}^{N} CE\Big(D_{img}\big(x_n, E_{text}(C_n)\big), M_n\Big). \quad (4)$$

More details can be found in the supplementary and code.

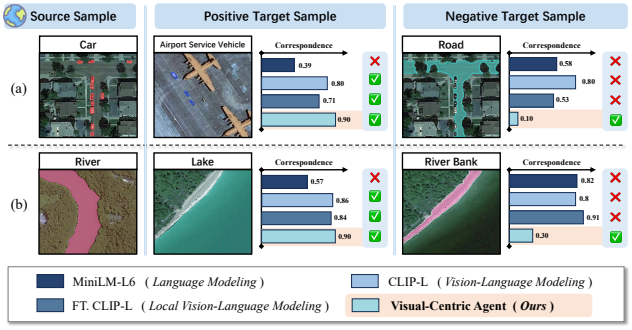


Figure 6. Positive and negative sampling mechanism according to the correspondence score between source sample and target sample. **(a):** Due to the contextual dependence in language modeling or vision-language modeling [20], "road" is mistakenly assigned as the positive sample of "car", while our agent employs the correct sampler. **(b):** Another case to further valid this point.

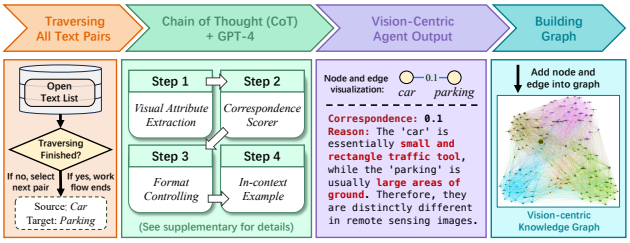


Figure 7. Constructing illustrations of the vision-centric knowledge graph by an elaborate vision-centric agent flow.

4.3. Vision-Centric Modeling

Challenge. Different from the previous close-set semantic segmentation dataset, the Sky-SA dataset is annotated in open-world texts. Although this manner provides more open vocabulary support, the ambiguous sampling challenge often leads to training collapse in discriminative models. As illustrated in Fig. 6, "car" is mistakenly regarded as the positive sample of "road" by computing the similarity between their text-embeddings due to their strong text-level or image-level contextual relationships.

Visual-Centric Knowledge Graph. To overcome the challenge, inspired by the in-depth exploration of visual representation by Xie *et al.* [22] and GraphRAG [81], we build a visual-centric agent using GPT-4 to obtain the visual relevance score for each text pair of the Sky-SA dataset. As shown in Fig. 7, the score is measured by the meticulously chain of thought about visual attributes, including appearance, color, and shape *etc.*. When all text pairs in Sky-SA are fully traversed, we can construct an offline vision-centric knowledge graph, which mainly consists of three elements: source node, target node, and edge. The edge indicates the visual-relevance score and reasons from source node to target node. Based on the graph, we perform flexible sampling for positive and negative samples by setting a sampling threshold illustrated in Algorithm 1.

Algorithm 1 Finetuning Stage for SkySense-O.

```
# Input (Given B=1): Image-I ∈ ℝ^{3×H×W}, Mask-M ∈ ℝ^{L×H×W},
# OpenTextList-O ∈ 𝕊^M, TextBeforeSample-T ∈ 𝕊^L,
# TextAfterSample-C ∈ 𝕊^N, Graph-G, Threshold-β (Default=0.5).

def TextSampler (T, G, O, C, β):
    for source_node in O_t:
        for target_node in O_t:
            if G(source_node, target_node) < β:
                neg[source_node] = target_node
    neg_sample = RandomSelect(T, neg, len(C)-len(T))
    text_batch = Concat(T, neg_sample) # [N,1]
def VisionCentricRegularization (T):
    S_t = sim(Traverse(Text_Embed(T))) # [N,N]
    S_v = G(Traverse(T)) # [N,N]
    L_vc = (1 − S_v) * S_t * L_region
def forward (I, T):
    I_feat = ImgEncoder(I)
    text_batch = TextSampler(T, G, O, C, β)
    Pred = VisualDecoder(I_feat, TextEncoder(text_batch))
```

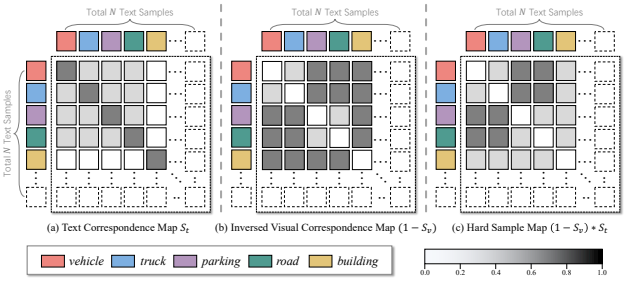


Figure 8. Sample similarity matrices in computing vision-centric regularization. Hard samples, such as *vehicle* and *road*, are exploited by multiplying S_t and inversed S_v.

Vision-Centric Regularization Loss. In RS imagery, the objects and land-cover are often densely packed and intricately intertwined, posing challenges for fine-grained visual-language modeling. To relieve this issue, we further exploit the potential of the vision-centric graph and design a vision-centric regularization loss, $\mathcal{L}_{\text{vc}}$. The regularization is built on the observation that the samples with strong contextual relationships but weak visual relevance scores are hard to distinguish as shown in Fig. 6. Specifically, given selected M negative samples C_m for class C_i, their textual similarities S_t are computed by their embedding similarities. Then, given text pairs, the visual relevance $S_v^{i,m}$ can be retrieved by the visual-centric graph. Finally, $\mathcal{L}_{\text{vc}}$ is computed as Eq. 5 and details are in Algorithm 1 and Fig. 8.

$$\mathcal{L}_{\text{vc}} = \frac{1}{BN} \sum_{i=1}^{B} \sum_{n=1}^{N} \left\| \left(1 - S_v^{i,m}\right) * S_t^{i,m} * \mathcal{L}_{\text{region}} \right\|^2. \quad (5)$$

Total Loss. In summary, the total loss for the finetuning stage of SkySense-O is defined as follows:

$$\mathcal{L}_{finetune} = \mathcal{L}_{pixel} + \lambda_1 \mathcal{L}_{region} + \lambda_2 \mathcal{L}_{vc}, \quad (6)$$

where λ_1 and λ_2 are two hyperparameters and set as 0.4 and 0.4 respectively for optimal ratio via performing ablations.

Model	Publication	Object Semantic Segmentation				Land-Cover Semantic Segmentation				
		iSAID [33]	SOTA [34]	FAST [34]	SIOR [34]	Potsdam [82]	OEM [30]	LoveDA [32]	Vaihingen	FloodNet [83]
Few-Shot Semantic Segmentation										
SegGPT [84]	ICCV 2023	29.17	31.41	19.55	37.47	30.12	25.19	31.40	47.27	33.81
DINOv [85]	CVPR 2024	25.35	18.24	7.92	25.04	37.65	23.05	25.17	35.10	44.55
Zero-Shot Semantic Segmentation										
SAN [80]	CVPR 2023	11.77	10.73	3.16	15.22	51.45	24.87	27.83	46.23	39.26
CAT-SEG [75]	CVPR 2024	20.58	13.71	3.13	19.64	50.81	30.26	37.15	49.62	41.36
SegEarth-OV [58]	Arxiv 2024	21.70	11.99	2.84	13.92	47.10	40.30	36.90	29.10	34.00
SkySense-O	-	**43.92**	**32.12**	**8.33**	**30.89**	**54.10**	**40.83**	**38.30**	**51.64**	**42.04**
	-	(+22.22)	(+18.41)	(+5.2)	(+11.25)	(+2.65)	(+0.53)	(+1.15)	(+2.02)	(+0.68)

Table 2. Zero-shot remote sensing object and land-cover semantic segmentation performances. Bold indicates the highest performance, and underline indicates the second highest performance in mIoU (%).

Model	Publication	SIRI-WHU [86]	AID-VQA [28]	RSVQA-HR [14]
		Acc.	Acc.	Avg. Acc.
GeoChat [35]	CVPR 2024	66.63	72.03	70.82
LHRS-Bot [87]	ECCV 2024	62.66	91.26	-
EarthGPT [88]	TGRS 2024	-	-	71.15
VHM [89]	Arxiv 2024	70.88	91.70	73.75
SkySense-O†	-	72.45 (+1.57)	92.95 (+1.25)	75.39 (+1.64)
SkySense-O	-	**74.79** (+3.91)	**94.10** (+2.40)	**78.09** (+4.34)

Table 3. Zero-Shot visual question-answering performances, compared with existing MLLMs. † indicates wo/ pixel-level fine-tuning by Sky-SA.

Model	Publication	RESISC [29]	AID [28]
CLIP [57]	PMLR 2021	69.25	66.70
RemoteCLIP [74]	TGRS 2024	70.85	74.3
CLIP-laion-RS [31]	-	71.70	71.21
SkyScript [31]	AAA1 2024	71.70	70.94
GeoRSCLIP [24]	TGRS 2023	73.90	76.27
SkySense-O†	-	**83.28**	**82.93**
	-	(+9.38)	(+6.66)

Table 4. Zero-shot remote sensing scene classification performances in Top-1 accuracy (%).

5. Experiment

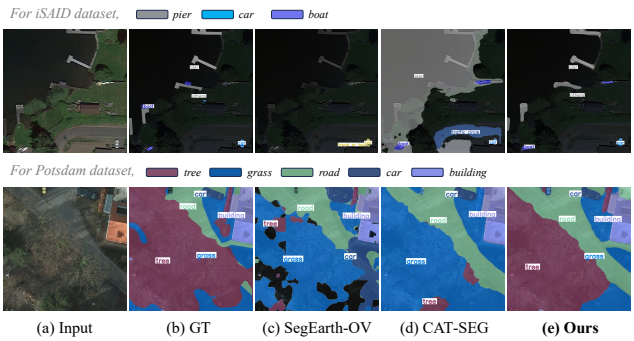
5.1. Evaluation Benchmarks

We evaluate SkySense-O on four representative RS interpretation tasks. To ensure fair zero-shot comparisons, all experiments are conducted without extra fine-tuning and labels from the evaluation datasets are not used during training. The supplementary materials provide further details on the implementation and visualization results.

Zero-shot object segmentation. We adopt iSAID [33], FAST [34], SIOR [34], SOTA [34] for the object segmentation experiment. They are chosen considering diverse scenes and categories. We report mIoU results on official validation sets. As depicted in Tab. 2, SkySense-O has achieved the SOTA performance on all object segmentation datasets. On average, it outperforms the previous SOTAs by an average margin of 14.27%. Furthermore, it even surpasses few-shot SOTAs on most datasets.

Zero-shot land-cover segmentation. We assess SkySense-O's land-cover segmentation performance on ISPRS Potsdam [82], Vaihingen and LoveDA [32] datasets. SkySense-O excels on all three datasets (Tab. 2). Notably, we surpass the second best CAT-SEG [75] by 1.41% mIoU and have achieved the best performance on the Vaihingen leaderboard. More importantly, our results demonstrate the potential to achieve more accurate interpretations than the ground truth (the second row in Fig. 9).

Zero-shot classification. We utilize two scene classification datasets: AID [28] and RESISC-45 [29]. The validation setting follows GeoRSCLIP [24] and Top-1 Accuracy is reported. SkySense-O overally outperforms competitive baselines, achieving 9.38% and 6.66% higher performance than state-of-the-art methods on the two datasets (Tab. 4).



(a) Input (b) GT (c) SegEarth-OV (d) CAT-SEG **(e) Ours**

Figure 9. Visualization of object segmentation results on iSAID dataset and land-cover segmentation results on Potsdam dataset.

Zero-shot RS VQA. To further validate the versatility of SkySense-O, we employ it in MLLM as a vision encoder for experimentation. Specifically, we maintain the same architecture as LLaVA-1.5 [90], except that we replace the original CLIP-L-14 [57] with our SkySense-O, and set the image resolution to 512×512 pixels. Regarding the setup of the instruction tuning data, during the pretraining stage, we utilize CC-558K dataset [60] and 121k image–text pairs converted from ChatearthNet [91] dataset. In the fine-tuning stage, based on instruction-tuning datasets collected by SkySenseGPT [61] from different sources, we select a subset of 593k data as our downstream instruction fine-tuning dataset, where details are in the supplementary. We evaluate zero-shot VQA performances on several public RS benchmarks as shown in Tab. 3.

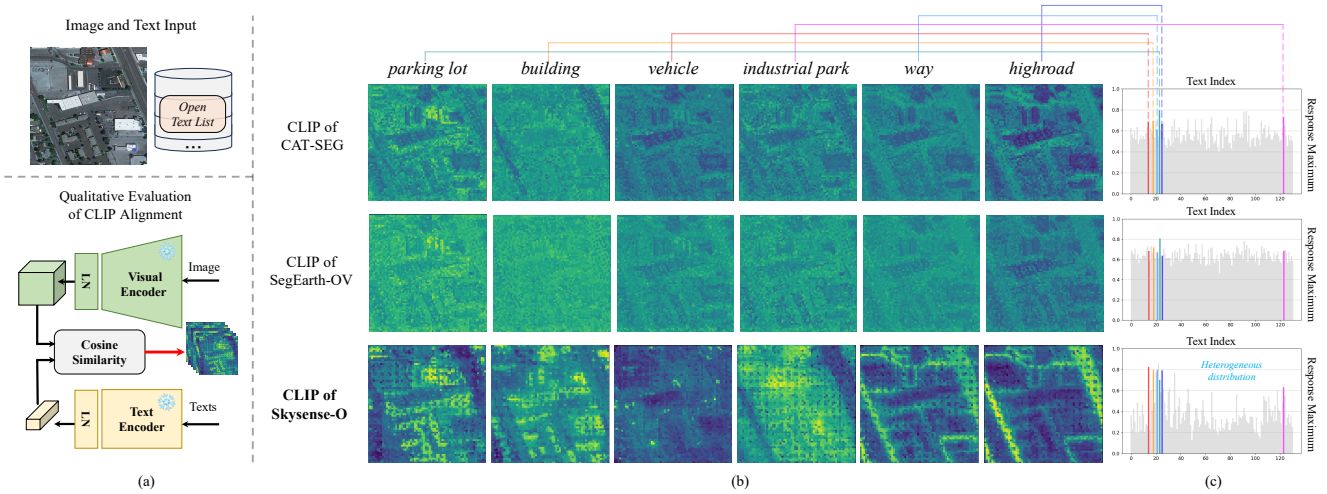


Figure 10. Visualization of text-image alignment in CLIP. (a) Computing the similarity between each text token and all image tokens to obtain text-image relevance (TIR) maps for different categories. (b) Visualizing TIR maps for some high-response texts. (c) Histogram of the maximum response values of TIR maps across the text list. The details of input texts can be referred to the supplementary materials.

Foundation Model	Vision Encoder	Vision-Enc. Parameter	Num. of Image-text	RESISC (Acc)
RemoteCLIP	ViT-L-14	304M	0.83M	70.85
SkyScript	ViT-L-14	304M	2.6M	71.70
GeoRSCLIP	ViT-H-14	630M	5M	73.90
Ours	**SwinV2-H-8**	**654M**	**10M**	**83.28**

Table 5. Comparison with existing RS vision-language foundation models. SkySense-O achieves consistent performance gains due to scaling laws in terms of model-size and data-size.

+Sky-SA	+Graph	+Pretrain-ing	+$\mathcal{L}_{vc}$	mIoU
				18.40
✓				27.76 (+9.36)
✓	✓			36.28 (+8.52)
✓	✓	✓		41.69 (+5.41)
✓	✓	✓	✓	43.92 (+2.24)

Table 6. Ablation study on the proposed dataset and designs.

5.2. Implementation Details.

In the pre-training stage, we use 80 A100-80GB GPUs and primarily adopt two large-scale remote sensing image-text datasets, SkyScript [31] and RS5M [24], and further expand their captions using Qwen-VL [92] and InternVL [93], comprising 10 million image-text pairs. In the fine-tuning stage, we employ OSM [25] for local image-text fine-tuning and Sky-SA for fine-grained interpretation capabilities on 8 A100-80GB GPUs as shown in Tab. 5. More details are included in the supplementary materials.

5.3. Discussions & Ablation Studies

Effectiveness of Image-Language Alignment. In Fig. 10, we follow MaskCLIP [94] and adopt the image-text correlation in the representation space to evaluate the effectiveness of vision-language foundation models. We select the CLIP models from SegEarth-OV [58] and CAT-Seg [75] for comparison. The former is a standard CLIP model, while the latter is a locally fine-tuned CLIP model by COCO-Stuff [19]. We randomly selected 131 category texts from Sky-SA as inputs. As shown in Figure 10(c), SkySense-O exhibits larger response variance across different categories, indicating stronger discriminative capabilities among different texts. In Fig. 10(b), we present several image-text correlation maps to further validate the effectiveness of SkySense-O in image-text alignment in open-world scenes.

- *vehicle vs. parking lot. (small object)* Because our modeling is vision-centric, small targets are effectively decoupled from the contextual background.
- *building vs. industrial park. (hypernym)* SkySense-O demonstrates proper alignment capabilities across different levels of semantic hierarchy due to large-scale remote sensing knowledge are learned in training.

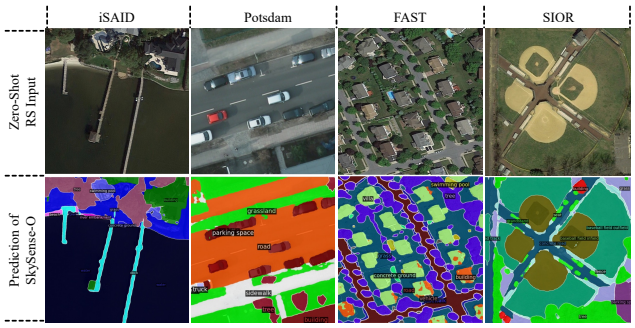


Figure 11. Visualizations of open-world interpretation results by SkySense-O on iSAID, Potsdam, FAST, and SIOR datasets.

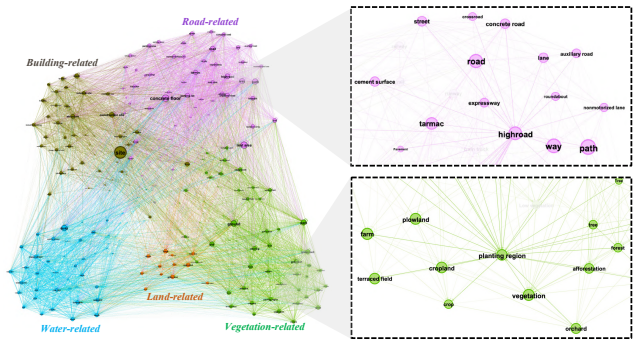


Figure 12. The visualization of vision-centric graph of the Sky-SA dataset. Node layout is performed via Force Atlas [95] and graph communities are detected by the Leiden algorithm [96]. Circles represent entity nodes with size proportional to their degree. Node colors represent entity communities. Five main communities are detected by an unsupervised manner and they just correspond to the major land-cover categories in the remote sensing imagery.



(a) Image (b) Wo/ vision-centric (c) W/ vision-centric

Figure 13. Comparisons of the results without and with vision-centric regularization loss. Compared to (b), the vehicles in (c) are clearly distinguished from the contextual floor.

Design of Vision-Centric Strategy. Tab. 6 presents the ablation study to assess our Sky-SA dataset and vision-centric designs, namely vision-centric graph sampler, pre-training foundation model, and vision-centric regularization loss. We utilize the iSAID dataset as evaluation benchmark and report mIoU metric on the official test set. Initially, we use the standard CLIP model [57] and fine-tune it on the COCO-Stuff dataset [19]. Compared with the baseline, the Sky-SA dataset leads to a notable improvement compared to COCO-Stuff dataset. Nonetheless, we believe that the Sky-SA dataset holds considerable untapped potential, which might be constrained by the intricate entanglement of open-world text labels as demonstrated in Sec. 4.1. To relieve this problem, we construct a vision-centric graph for all text labels of the Sky-SA dataset as shown in Fig. 12. It is intriguing that the nodes in the graph naturally exhibit proper unsupervised clustering results by community detection [96], without using any text labels. This further validates the rationale for vision-centric modeling and the introducing of vision-centric graph leads to 8.52% mIoU improvement. Then, we replace the standard CLIP with our pre-trained model incorporating extensive remote sensing knowledge. This effectively improves the performance to 41.69% mIoU, validating the necessity of remote sensing pre-training model. Finally, we introduce vision-centric regularization loss, which uncovers and prioritizes hard samples in visual-language modeling. The results show that the constraint pushes the performance to a very strong 43.92% mIoU. Furthermore, as shown in Fig. 13, this design enhances the discrimination ability, especially between high-context categories.

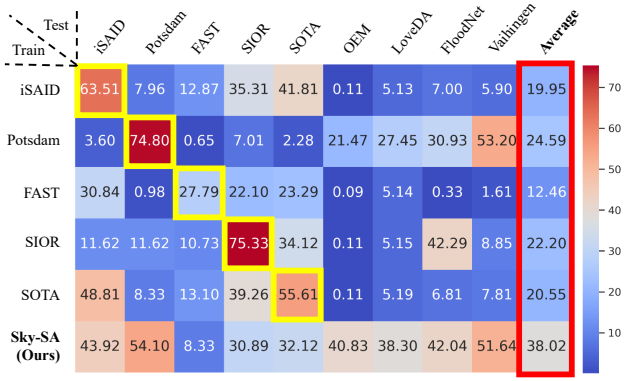


Figure 14. Zero-shot semantic segmentation results of Sky-SA and existing datasets. The vertical axis represents the training set, while the horizontal axis represents the test set. The yellow box denotes the result is fine-tuned by its own training set (*i.e. supervised learning*) and red box denotes the average performance.

Significance of Sky-SA for Open-World Interpretation. In Fig. 14, we replace Sky-SA with existing datasets and conduct experiments under the same settings as in Tab. 2. It can be observed that existing datasets exhibit poor generalization to other datasets, except on their own test sets. In contrast, the Sky-SA dataset achieves superior zero-shot segmentation performance across all datasets, outperforming the second best dataset by an average of over 13.43% mIoU, which demonstrates the significant potential of Sky-SA for open-world interpretation.

6. Conclusion

In this paper, we present Skysense-O, a large-scale and general remote sensing interpretation model for open remote sensing images. Skysense-O is mainly driven by the proposed semantically-rich and fine-grained dataset Sky-SA and the vision-centric visual-language modeling, consistently outperforming other models on a variety of tasks, showcasing its exceptional generalization ability and strong performance. We hope this work will constructively push the RS open-world interpretation on constructing large-scale vision-language datasets and models for further researches. Our future work aims to explore fine-grained remote sensing interpretation, such as analyzing object states and relationships, further enhancing the flexibility of open-world applications.

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